SYNAPSYS

User manual, version 0.1

Characterisation tool v5.22

The launcher

SYNAPSYS is started from `hub.py`, or from the hub executable in the installed version. The hub is a single window with three tiles, one per module, and has nothing to configure (Figure 1). Each button opens its module in a window of its own, so the hub can stay open and the same module can be launched more than once.

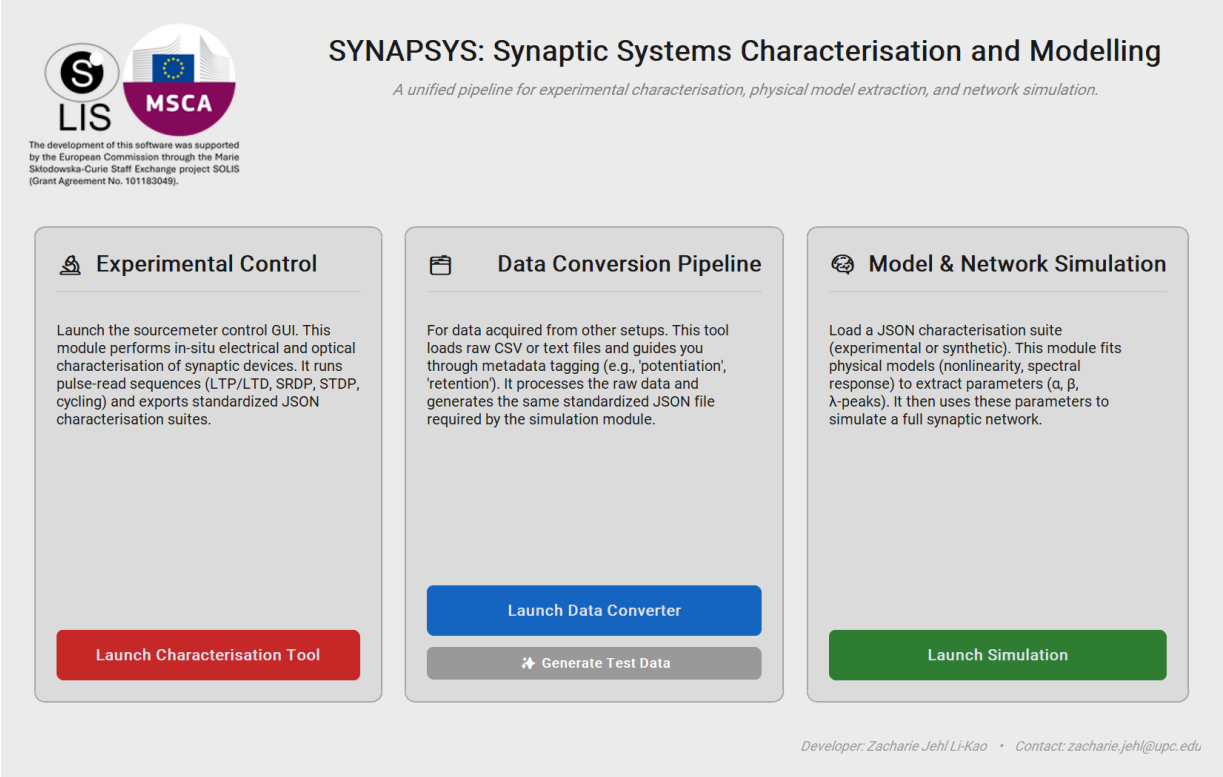


Figure 1. The hub. Red opens the characterisation tool, blue the data converter, green the modelling module.

The red tile opens the characterisation tool, which drives the Keithley sourcemeter. The blue tile opens the data converter, for measurements taken on another setup. The green tile opens the modelling module, which reads a characterisation suite and fits it. The modules are linked by files only: the characterisation tool and the data converter both write a JSON suite, and the modelling module reads it.

The grey button under the data converter, Generate Test Data, asks for an output folder and writes a set of synthetic measurements as CSV files, together with a `METADATA_INSTRUCTIONS.txt` that lists what to type in the data converter for each file. If no folder is chosen, the files go to `test_data_output` in the working directory. This is the quickest way to try the data converter and the modelling module without a device. The message box that appears after the click only confirms that the generator was started; the files are there once the folder has been chosen.

The characterisation tool

The characterisation tool drives a Keithley 2600-series sourcemeter and measures the conductance of a synaptic device under electrical or optical stimulation. The window is divided in three parts (Figure 2 shows it after a first run). The left panel holds the parameters and scrolls, the centre shows the stimulus and read waveforms above the measured curve, and the right panel gives a numerical summary of the last run, the list of runs kept in memory and the logos. All the figures of this chapter were recorded against a simulated 2636B, so the values only illustrate what the screens look like.

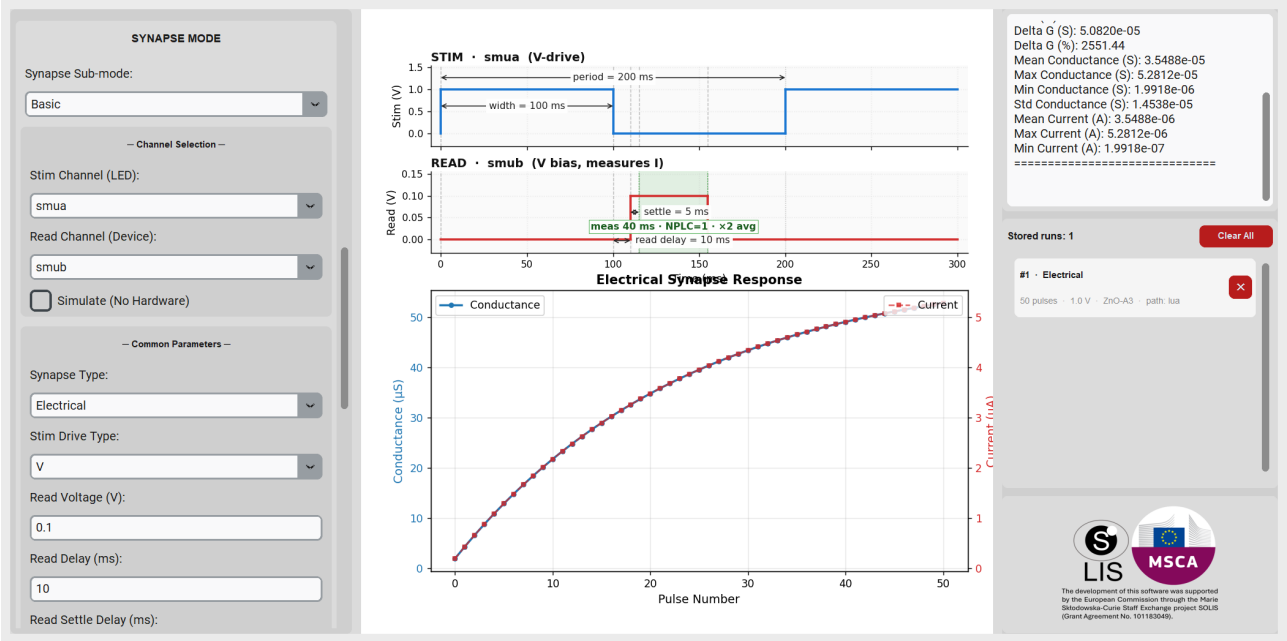


Figure 2. The window after a first Basic run. The left panel holds the parameters, the centre the pulse schematic and the measured curve, the right panel the summary and the stored runs.

The connection is set in the General Parameters at the top of the left panel (Figure 3). Connection Type is GPIB, RS232 or LAN, and Port Address is the VISA address for GPIB and RS232, or the IP address of the instrument for LAN. There is no Connect button: the instrument is opened when a measurement is started, identified, and released at the end of the run. A model that cannot be identified stops the run, because the voltage limit of the safety checks depends on the model. Measurement Speed is the NPLC of the read; a larger value gives a quieter reading and a longer read window. Mains Frequency is detected from the instrument by default and should only be set by hand to 50 or 60 Hz if the detection is not wanted, since the NPLC only rejects mains pickup when it matches the grid. The sample identifier of each sub-mode is stored with every run and ends up in the exported files.

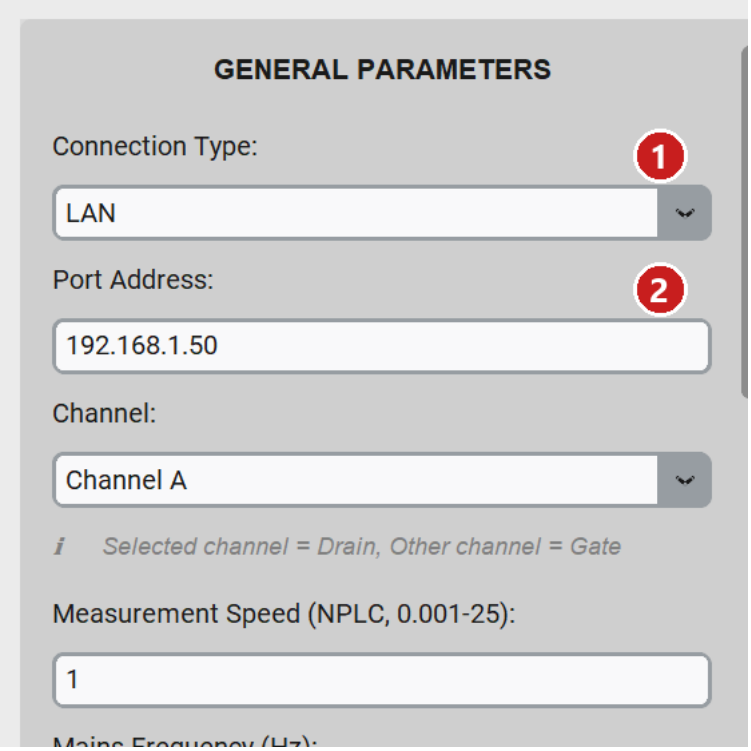


Figure 3. Connection type (1) and port address (2).

Measurement Mode, below the general parameters, selects Diode, Transistor or Synapse. The two first modes record current-voltage curves and are not covered here. In Synapse mode the Synapse Sub-mode lists Basic, Visual (Self-Powered), SRDP, STDP, Cycle, Multi-Device and Retention, and the parameters below change with the choice (Figure 4). Stim Channel is the source that applies the stimulus (an LED or an electrical pulse) and Read Channel is the one that measures the device. The checkbox Simulate (No Hardware) allows to try the interface without an instrument.

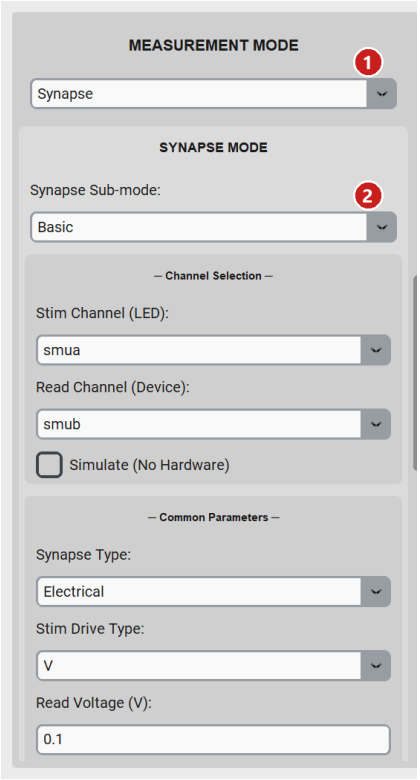


Figure 4. Measurement mode (1) and synapse sub-mode (2).

Basic, SRDP and STDP share a block of common parameters. Synapse Type is Electrical, Visual or Memristor (Pulse), and Stim Drive Type is voltage or current. The read is made at Read Voltage, starting Read Delay after the end of the stimulus pulse, after a Read Settle Delay that lets the current range settle, and during Read Pulse Width. The read settle delay matters when the device is measured on a low current range: the range needs about 2.5 ms to settle at 1 µA, 15 ms at 100 nA and 90 ms at 10 nA, and a shorter value is reported as a settling warning at the end of the run, because the first readings are then still drifting. Samples per Pulse set to 0 lets the tool choose.

The Basic sub-mode applies a train of identical pulses and reads the conductance after each of them. The fields are Stim Level, Stim Width, Stim Period, the number of pulses, and the wavelength and intensity of the light source, which is set by hand and only recorded with the run. The presets fill typical sets of values. Run Measurement starts the train. The plot shows the stimulus and the read waveforms of the first period, with the width, period, settle time and read delay dimensioned, and below them the conductance and the current at each pulse. Point 0 is the baseline measured before the first pulse, so that a train of n pulses gives n + 1 points. When the run is finished a message reports where the train ran and whether the period was kept (Figure 5). Execution path lua means that the train was run by the instrument itself, which is the accurate case, and standard means that it was timed by the computer, which is less precise for short periods.

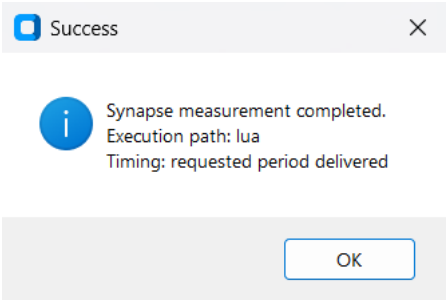


Figure 5. The message at the end of a run.

Visual (Self-Powered) is for devices that produce a photocurrent without bias. The LED is driven with Light Pulse Voltage during Light Pulse Width every Light Pulse Period, and the current is read at 0 V during the pulse, between Start Delay and End Margin (Figure 6). A Light Pulse Voltage above 2.5 V has to be confirmed before the run starts. The checkbox Continuous I(t) Mode records the current continuously at every Sample Interval, whereas the default gives one value per pulse. The plot gives the photocurrent at each pulse. A Visual run cannot be selected for a suite, since a photocurrent is not a conductance.

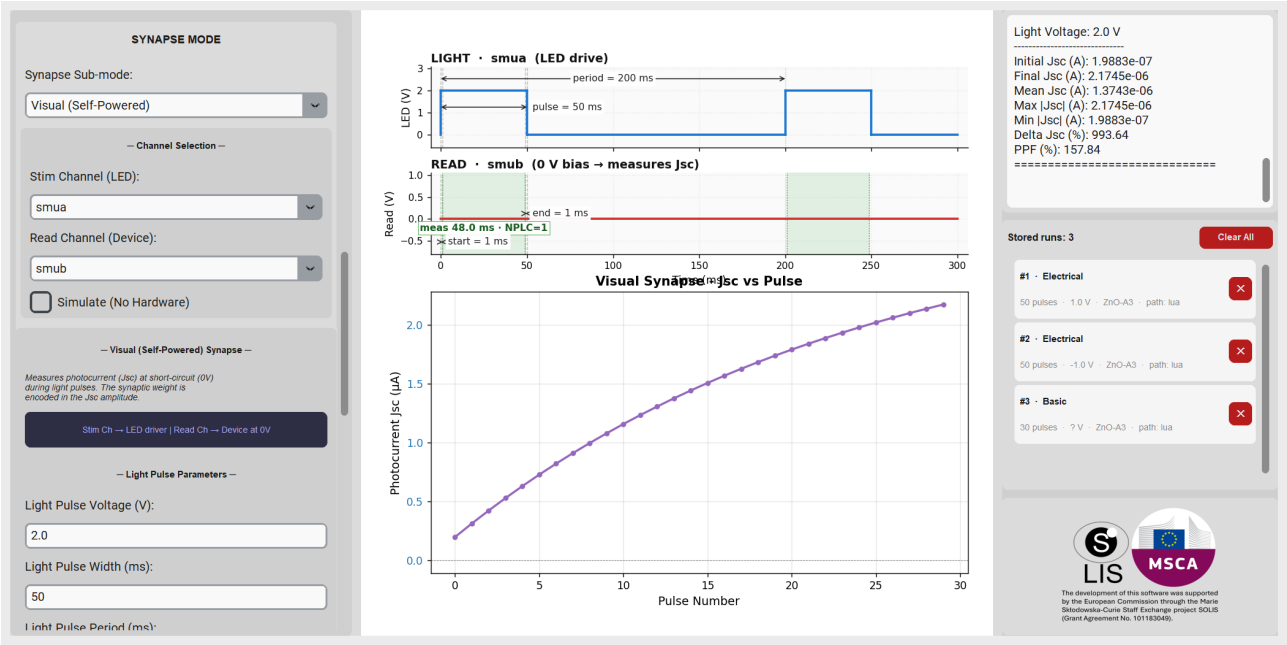


Figure 6. A Visual run. The photocurrent increases pulse after pulse.

SRDP measures the change of conductance as a function of the pulse frequency. The frequency goes from Freq Start to Freq End in the given number of points, on a linear or a logarithmic scale, with a given number of pulses at each frequency (Figure 7). The highest frequency that the instrument can deliver is set by the stimulus width, the read delay and the measurement time. When the requested range goes beyond it, a message gives the limit and asks whether to continue (Figure 8). Continuing runs the points above the limit as fast as the instrument can, and the frequency in the file is the one that was delivered. After such a run the end message reports that the period was not delivered (Figure 9). The data are valid, however, and a shorter stimulus width or a smaller NPLC raises the limit if the requested frequencies are needed.

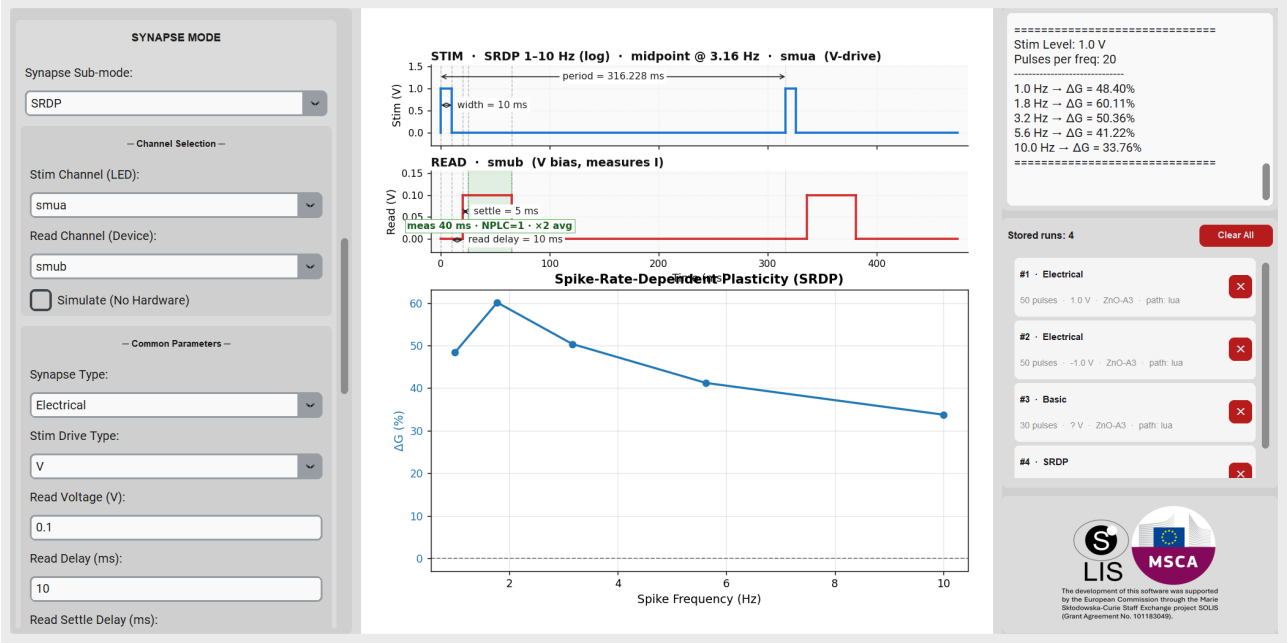


Figure 7. An SRDP run.

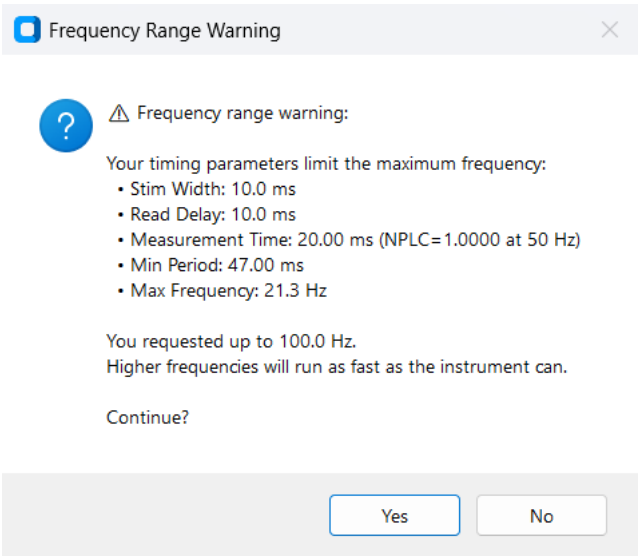


Figure 8. The frequency range message.

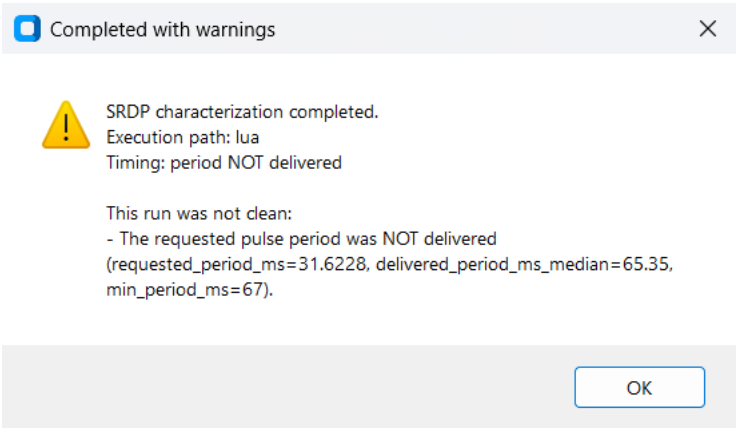


Figure 9. End of a run in which the period was not delivered.

STDP applies a pre-spike on the stimulus channel and a post-spike on the read channel, separated by Δt, and gives the change of conductance as a function of Δt (Figure 10). The fields are the range of Δt and its number of points, the number of spike pairs, the pre and post levels, the pulse width and the pair period. A range with a point at Δt = 0 is refused, since there is no order between the two spikes, and an even number of points on a symmetric range avoids it.

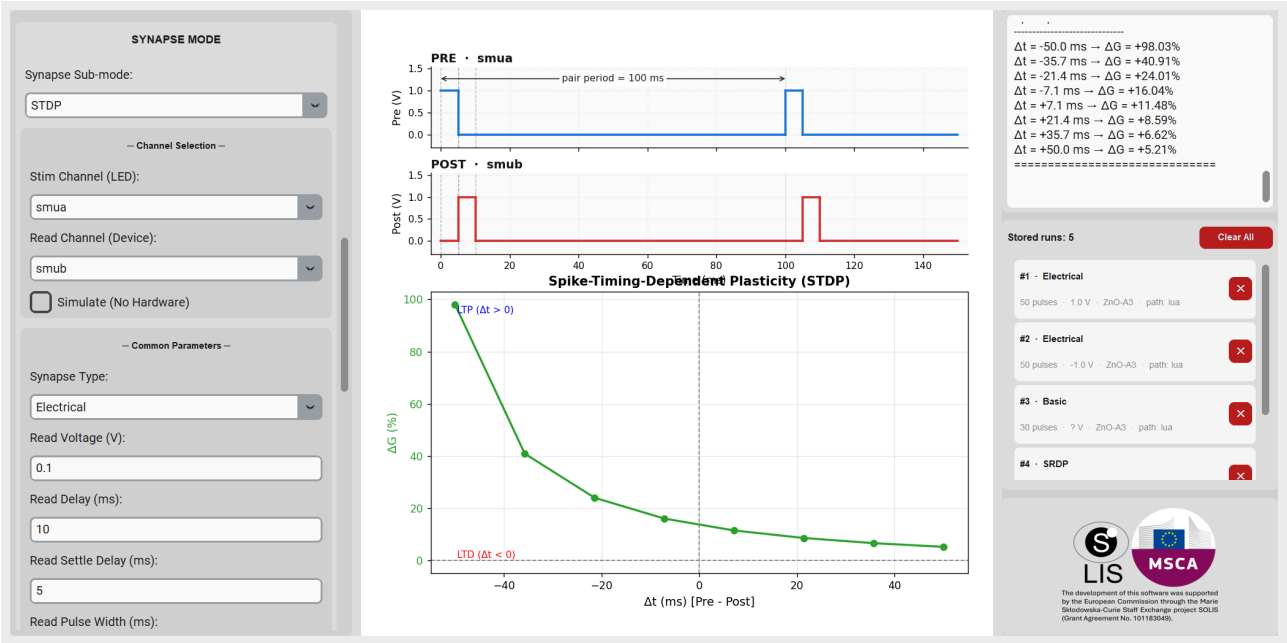


Figure 10. An STDP run.

Cycle alternates two trains, A and B, and repeats them for the chosen number of cycles. Train A potentiates and train B depresses with the default values. Each train has its own channel mode (a single SMU that writes and reads, or two SMUs, one for each), write level and width, read voltage, period and number of pulses, and train B can be disabled. # Cycles, Inter-train Delay and Inter-cycle Delay set the rest of the sequence. The plot shows the conductance against the pulse number for all the cycles (Figure 11), and the summary gives the mean change of conductance in each phase, its reproducibility, the dynamic range and the ON/OFF ratio. Multi-Device uses the same parameters and repeats the cycle sequence on a number of devices one after the other, with a delay between devices and a name pattern that numbers the samples.

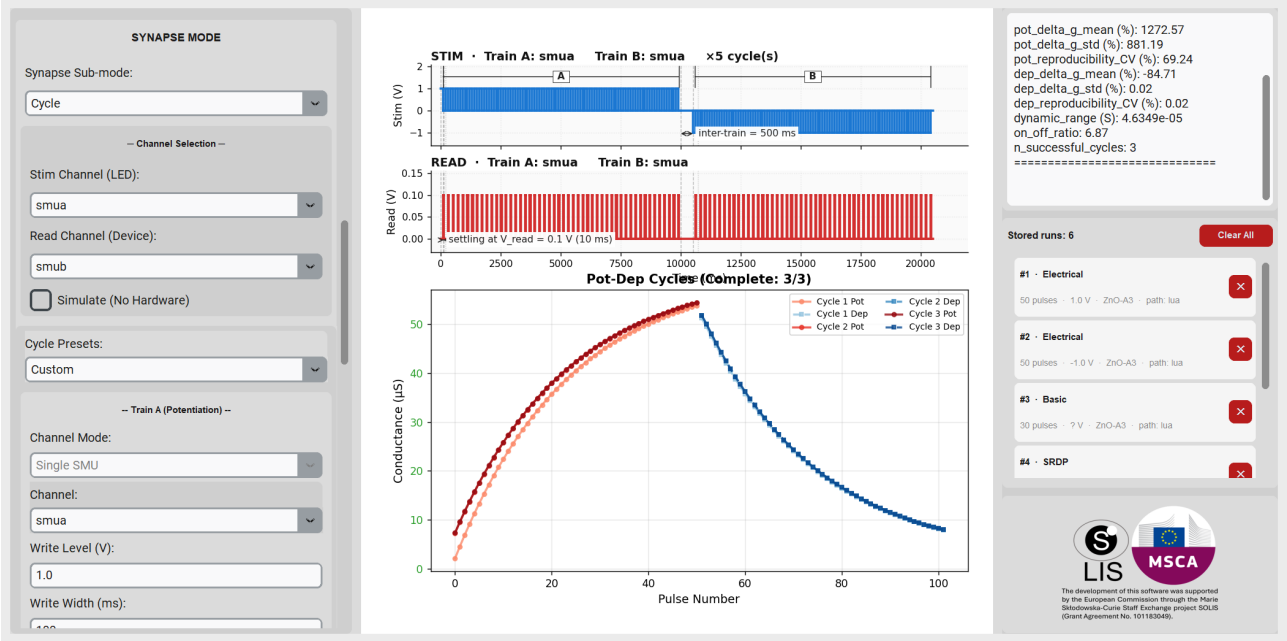


Figure 11. A Cycle run of three cycles.

Retention follows the conductance of a device after it has been set to a state. The device is first potentiated with another sub-mode and the retention run is started afterwards. The conductance is read at Read Voltage, either at the sample times given as a comma-separated list in seconds, or at every Interval between readings for the given number of points (Figure 12). The run takes as long as its schedule, since it

is timed by the computer.

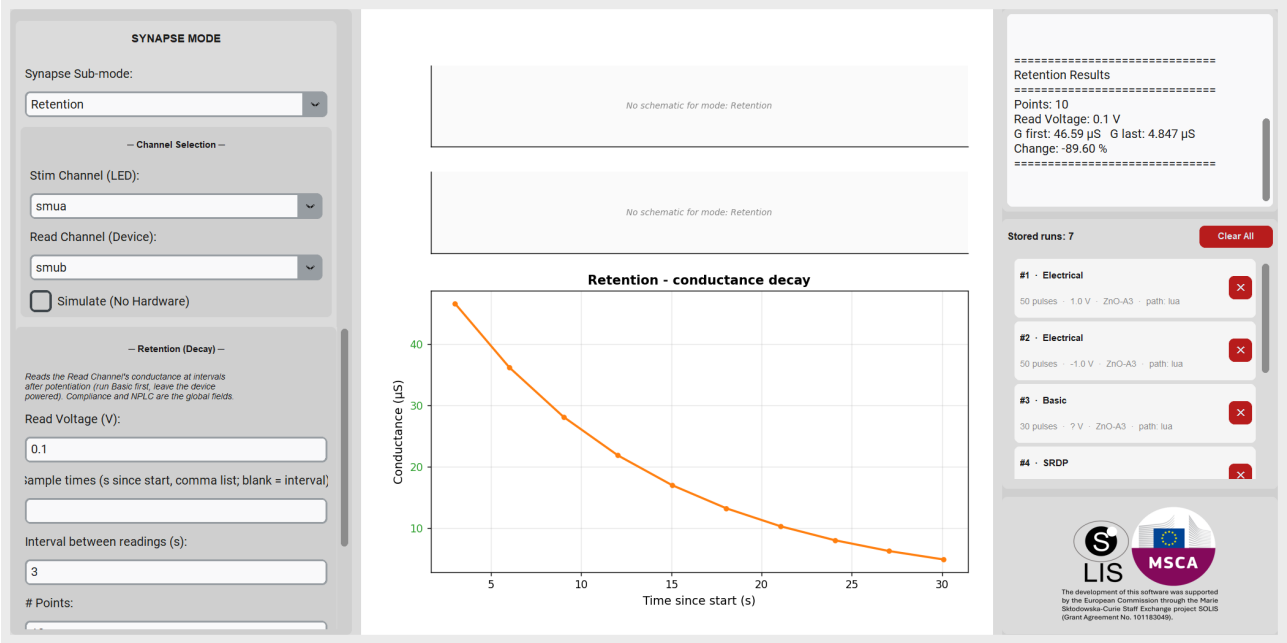


Figure 12. A Retention run. The conductance decays after the potentiation.

Every run is kept in memory and appears in the Stored runs list of the right panel, where its cross removes it and Clear All empties the list. Save Data writes the displayed run to a CSV file, and the header of the file records the execution path of the run. A run that is neither saved nor exported is lost when the tool is closed. Export Characterization Suite, at the bottom of the Cycle and Retention panels, assembles the stored runs in the JSON file that the modelling module reads. The window lists the stored runs, and each run to be used is selected and given its role with the Mark as buttons: LTP for the exponent α, LTD for β, Cycles for the dynamic range, Retention for the decay time constant, STDP and SRDP (Figure 13). Optical runs at three wavelengths or more are recognised as a wavelength sweep without being marked, and every run keeps its own wavelength and intensity. Export Suite then asks for the file name. The suite is checked before it is written, and a missing or invalid element is listed in a message. A suite that lacks a measurement is accepted, and the corresponding parameters are then shown as defaults by the modelling module.

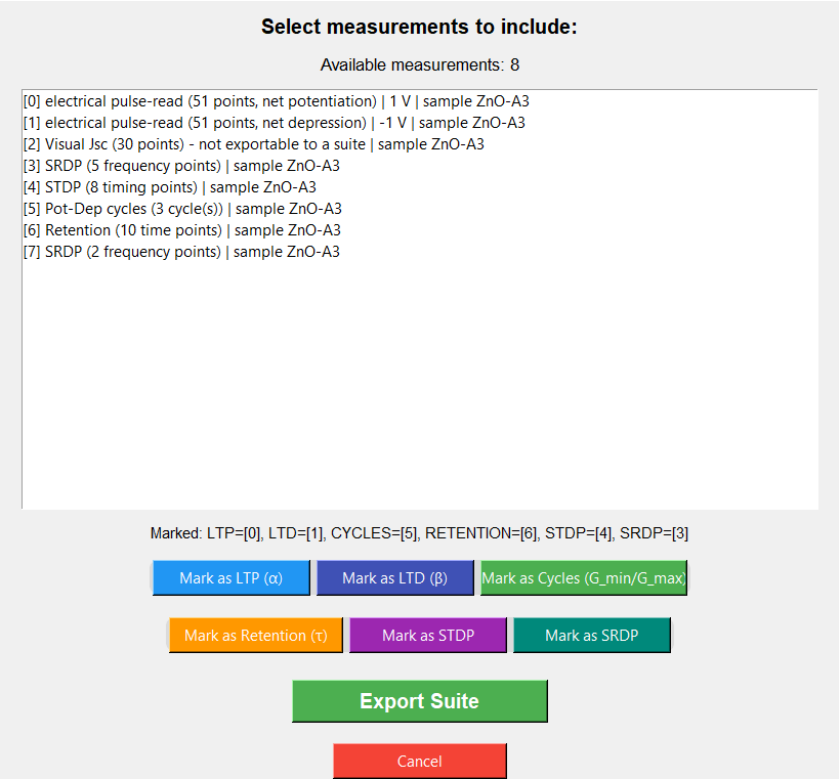


Figure 13. The export window with the roles marked. The list shows the stored runs in the order in which they were made.

The data converter

The data converter turns measurements made on another setup into the same JSON suite as the one that the characterisation tool exports. A file has to be a CSV, Excel or text table with at least two columns, the first for the x axis and the second for the y axis, and every other column is ignored. Each file is loaded with a description of its content, called its metadata, because the modelling module uses it to know which model parameter the file informs.

There are two ways to load files. Load Data File takes one file and opens the metadata window. Batch Load (with metadata file) first asks for a text file with the metadata of every file and then for the folder in which the data files are (Figure 14). The text file has the format of the `METADATA_INSTRUCTIONS.txt` written by Generate Test Data, which is the easiest way to see how it is built. Each dataset appears in the list on the left with its type, axes, number of points and change of conductance, which are computed when the file is loaded.

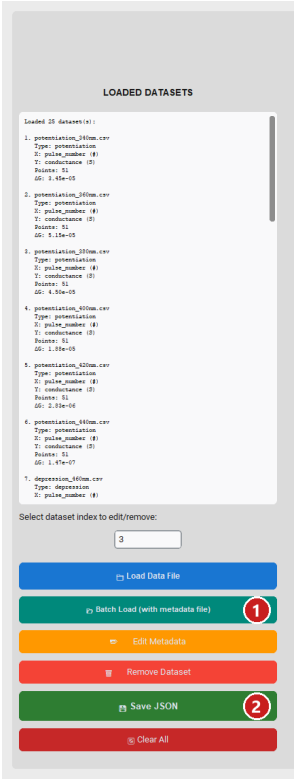


Figure 14. The converter after a batch load of the test data. The numbers mark Batch Load (1) and Save JSON (2).

The metadata window asks what the axes represent and their units, and the type of experiment, for instance potentiation, depression, retention, SRDP or STDP (Figure 15). The fields for the stimulus change with the type: the wavelength and intensity of the light, or the voltage of an electrical pulse, then the pulse width, the frequency or period, the number of pulses and the read timing, and the cycle or train parameters when the type needs them. Missing readings in a file are kept as gaps and are reported by the modelling module. The OK and Cancel buttons are above the stimulus parameters in the window, and the stimulus fields have to be filled before OK is pressed. Each wavelength of a sweep is its own file with its own wavelength and intensity.

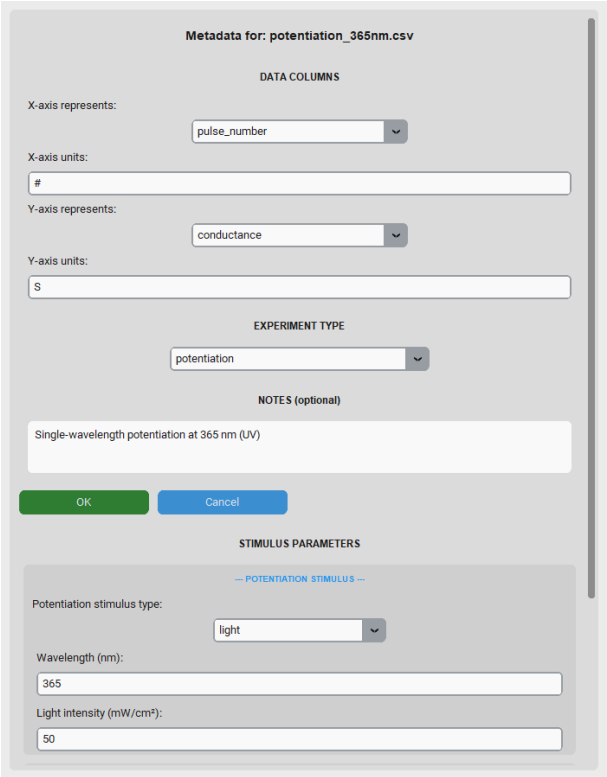


Figure 15. The metadata window for a potentiation measurement at 365 nm.

A dataset is corrected with Edit Metadata after its index has been typed in the field above the buttons, and removed with Remove Dataset. Clear All empties the list. Save JSON writes the suite, with a file name that carries the date and time by default.

The modelling module

The modelling module simulates a synaptic device with a physical model, extracts the parameters of that model from a characterisation suite, and uses them to simulate arrays of devices and spiking networks. The window has a left panel, called Synapse Controls, that is shared by all the tabs, and four tabs: Simulation, Model Fitting, Network and SNN. Before any file is loaded the simulated device uses a set of default parameters, and the box Synapse State at the bottom of the left panel says whether the parameters are those defaults or come from a loaded file.

The Simulation tab shows the conductance of a single device while pulses are applied (Figure 16). The Operation Mode is Potentiation (LTP) or Depression (LTD), and a depression can be driven optically or by voltage pulses. The sliders set the intensity (0.1 to 100 mW/cm^2), the wavelength, the pulse width (10 to 500 ms), the rest between pulses and the number of pulses (1 to 50). The wavelength range follows the fitted spectral response when a model is loaded. Apply Potentiation, which reads Apply Depression in depression mode, runs the pulse train and draws the conductance above the wavelength response curve of the device, with the light pulses as a dashed line and the current wavelength marked on the curve. When a fitted spectral response is loaded, the wavelength decides if a pulse potentiates or depresses (a positive lobe of the curve potentiates and a negative lobe depresses), so the requested operation mode cannot override it. The Synapse State box reports the branch that was taken when it differs from the one that was requested. Auto Demo runs a fixed sequence at several wavelengths, and Reset brings the device back to its initial state.

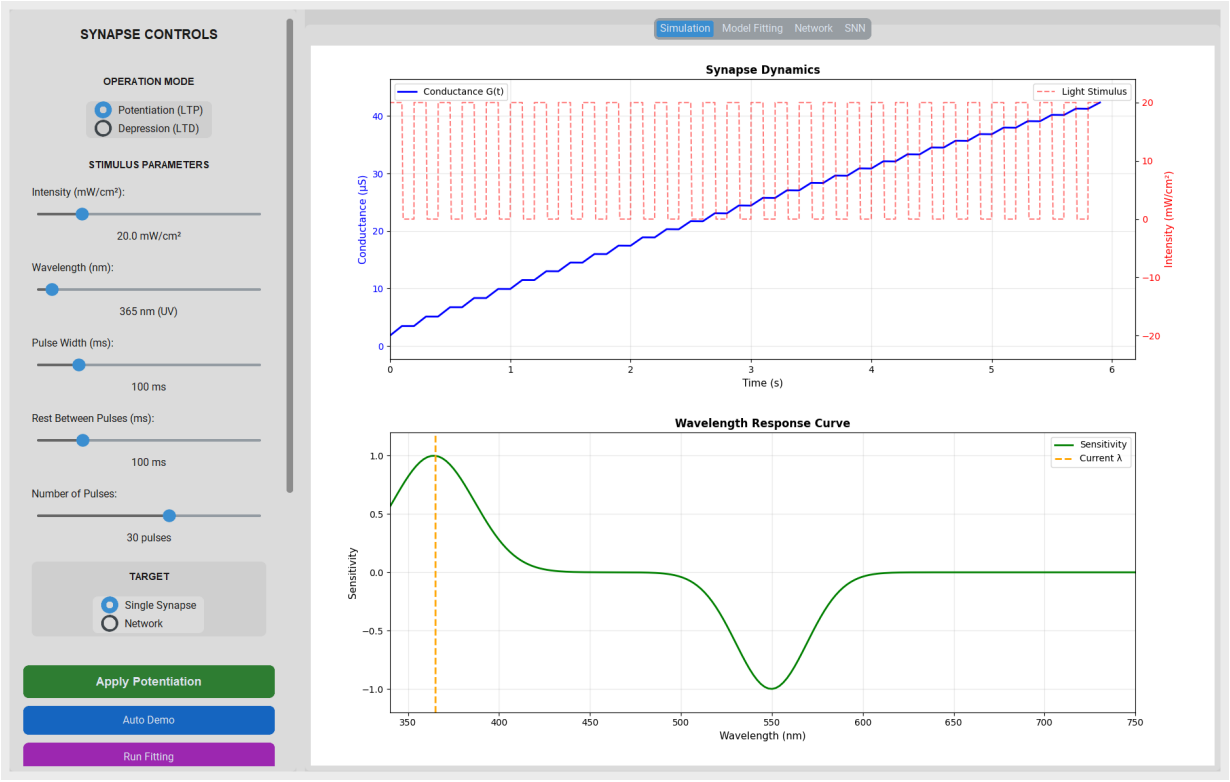


Figure 16. The Simulation tab after thirty pulses at 365 nm with a loaded model.

The Model Fitting tab starts from a characterisation suite. Load Experimental Model opens a JSON file, which is either a suite written by the characterisation tool or the data converter, or a fitted model saved earlier. For a suite the parameters are extracted at once, and the text lists them together with the quality of each fit (Figure 17). The parameters are G_min, G_max and their ratio, the exponents α and β of the potentiation and depression nonlinearities, the rates A_peak and B_peak, the peak wavelength and width of the spectral response, and the retention time constant. A parameter that could not be extracted from

the file is not hidden: it is followed by a tag that says that it is missing from the file, that its fit failed, or that it is a default and was not measured, and a block at the end of the text lists everything that was not measured. A fit with R^2 below 0.8 is tagged POOR. A_peak and B_peak depend on α, whose uncertainty is reported with them, and the rates k_A and k_B printed under them do not, so that devices are better compared on k. Save Fitted Model writes the model to a JSON file that Load Experimental Model can read again.

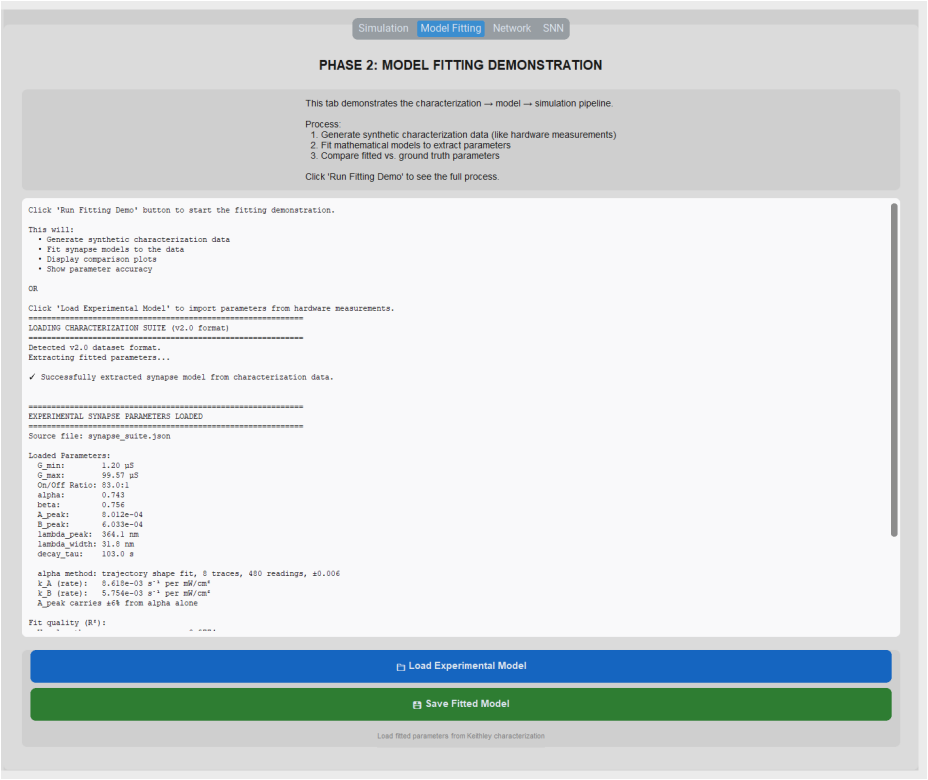


Figure 17. The Model Fitting tab after loading the suite exported by the data converter.

Once a model is loaded, the button Run Fitting Demo is renamed Run Fitting. It first asks about the number of peaks of the wavelength response when this was fitted (Figure 18). The dialog gives the number of peaks that was detected, with their wavelength and their type, and the fit quality. Use automatic keeps the result, and forcing one, two or three peaks refits the spectral response with that number. A forced refit replaces the simulated device, and a network or an SNN that exists then has to be initialised again. The fitting window that follows draws the data and the fit for the wavelength response, the nonlinearity, the retention, the dynamic range, the STDP window and the SRDP response (Figure 19). It has Save Plot and Close Window buttons, and it can be enlarged. Without a loaded model, Run Fitting Demo generates a synthetic suite, fits it and compares the fitted values with the values used to generate it. It does not change the simulated device.

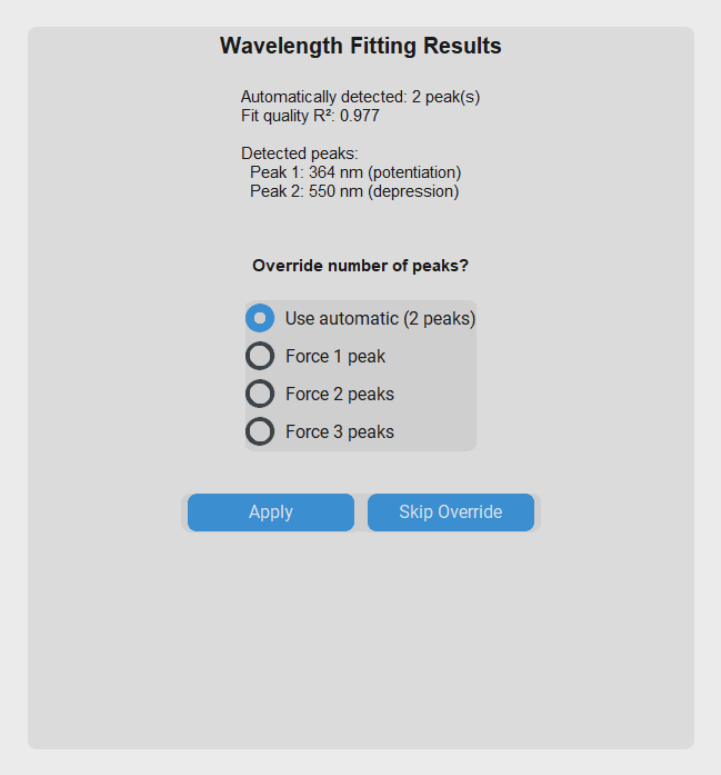


Figure 18. The question about the number of peaks.

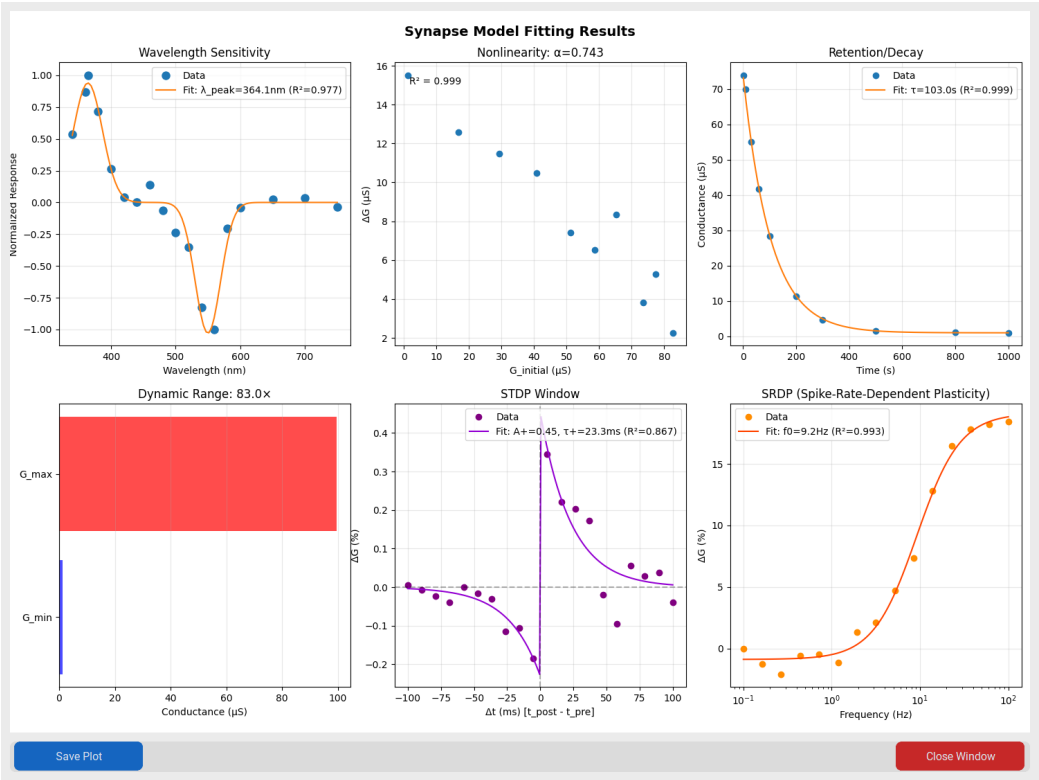


Figure 19. The fitting window. Each panel shows the measured points and the fitted model.

The Network tab simulates an array of devices. Choosing Network as the Target in the left panel adds the size of the array (rows, columns, with quick buttons from 3 × 3 to 10 × 10), the device variability in per cent and a list of spatial patterns: bars, diagonals, a cross, corners, a checkerboard, a circle, a duck, a thumb up, a raised fist and others. Load Excel Pattern adds a custom pattern from an Excel file, which is then resized to the array. The network mode is rate-based, which applies the patterns to the devices directly, or spiking,

in which the synapses learn with STDP. Changing the mode discards the existing network. Initialize Network builds the array with the loaded parameters, and with the variability given in the left panel, so that every device is slightly different. The Apply button of the left panel, which reads Apply Pattern to Network when the target is the network, then applies the selected pattern for the chosen number of pulses with the rest between them, and the tab shows the conductance of every device next to the intensity of the last pattern (Figure 20). If no network exists the tool offers to initialise one. Rest (500ms) lets the array relax, Show Evolution opens the window with the successive states (Figure 21), and Reset Network brings the array back to its initial state. Run Network Demo initialises an array and applies four patterns of five pulses each at the potentiation peak of the loaded model.

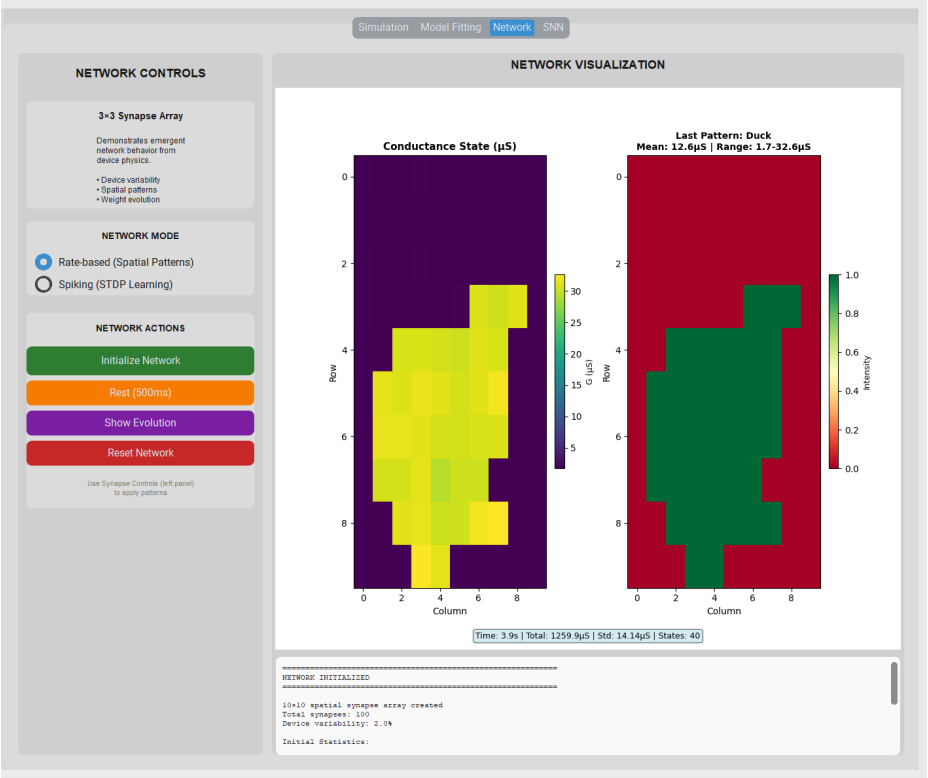


Figure 20. The Network tab after twenty pulses of the pattern Duck on a 10 × 10 array at 365 nm.



Figure 21. The window opened by Show Evolution.

The SNN tab simulates a spiking network with an input layer and a hidden layer, whose synapses are the devices of the model (Figure 22). The fields are the number of input neurons, the number of hidden neurons, the input firing rate and the duration of the simulation, and the checkbox Enable STDP Learning switches the weight update on. The input spikes are drawn from a Poisson process. Run SNN Simulation prints the setup in the output box and draws the input spikes, the hidden spikes and the weights before and after the run. The STDP parameters are the fitted ones when the loaded suite contains an STDP measurement, and the documented defaults otherwise, and the output box says which of the two is used. A warning is written in the output box when the number of input neurons is too small for the device variability. Reset SNN clears the network, its results and the output.

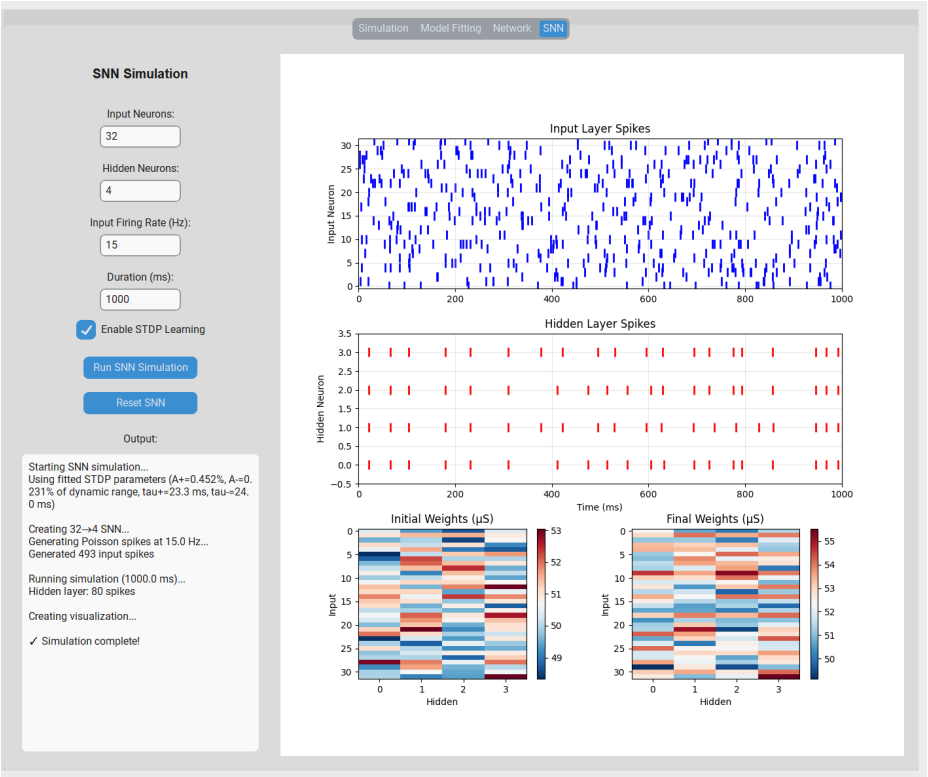


Figure 22. The SNN tab after one second of simulation.